

FULL-STACK PORTFOLIO PROJECT

Real-Time Chat Application

using Spring Boot, WebSocket, React & PostgreSQL

React.js

Spring Boot

WebSocket

PostgreSQL

JWT

Email OTP

A secure real-time chat web application with user registration, email verification, login, online user presence, direct messaging, media messages, profile settings, theme selection, group chat, admin controls and polls.

Corrected tech stack version - no MERN / Node.js / Express.js / MongoDB / Socket.IO wording

What is this project?

This is a full-stack real-time chat application built using React.js frontend and Spring Boot backend. WebSocket is used for instant message delivery and online presence updates. PostgreSQL stores user accounts, chats, groups, messages, polls and profile data. JWT authentication secures user sessions.

Registration and login flow

Users sign up by entering username, display name, email and password. After submitting, an OTP is sent to the entered email and the user is redirected to the Verify OTP page. After successful verification, the account becomes active and the user can log in securely using JWT.

Real-time messaging

- Direct user-to-user chat
- Instant WebSocket updates
- Message history
- Edit and delete messages

Rich chat features

- Text messages
- Emojis and mentions
- Photo and video sharing
- Online/offline status

Profile and theme

- Update display name
- Upload profile picture
- Select chat themes
- Secure sign out

Group communication

- Create groups
- Assign admins
- Remove members
- Group settings and members

Polls and collaboration

- Create polls inside groups
- Group member participation
- Navigate to user inboxes
- Organized chat workspace

Security and backend

- JWT authentication
- Email OTP verification
- Spring Boot REST APIs
- PostgreSQL database

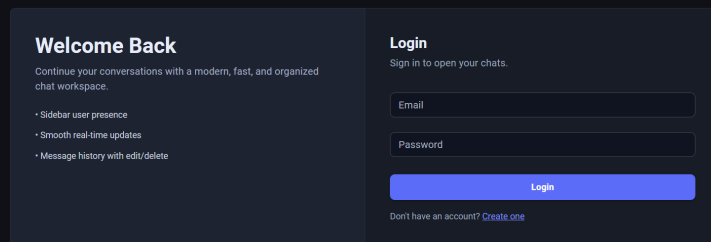
The screenshot displays three distinct authentication screens for a chat application, each with a dark theme and light-colored text and input fields.

- Login Screen:** Features the title "Login" and the instruction "Continue to your chat workspace." It includes input fields for "Username" and "Password", a "Show Password" toggle, a prominent green "Login" button, a "Forgot Password?" link, and a "No account? Create one" link.
- Register Screen:** Features the title "Register" and the instruction "Create your account and start chatting." It includes input fields for "Display Name", "Username", "Email", "Password", and "Confirm Password", each with a "Show" toggle for password fields. It features a prominent green "Create Account" button and a link for "Already have an account? Login".
- Verify Email Screen:** Features the title "Verify Email" and the instruction "Enter the OTP sent to your email." It includes input fields for "Email" (pre-filled with "anuradhawork123@gmail.com") and "OTP", a "Show OTP" toggle, a prominent green "Verify" button, a "Resend OTP" button, and a "Back to Login" link.

Authentication Flow - Login, Register and Verify Email

This screenshot explains the main account flow. Users can log in, create a new account, and verify their email using OTP before accessing the chat workspace.

Key items: Login page, Register page, Email OTP verification



The login screen features a dark background with a light gray card. On the left, a 'Welcome Back' section includes a sub-header, a descriptive paragraph, and a bulleted list of features. On the right, the 'Login' section contains a sub-header, a sign-in instruction, two input fields for email and password, a blue 'Login' button, and a link for new users.

Welcome Back

Continue your conversations with a modern, fast, and organized chat workspace.

- Sidebar user presence
- Smooth real-time updates
- Message history with edit/delete

Login

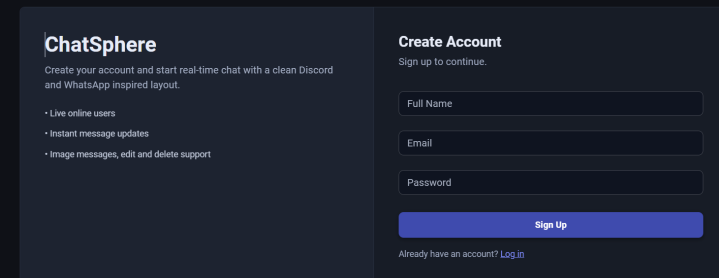
Sign in to open your chats.

Email

Password

[Login](#)

Don't have an account? [Create one](#)



The sign-up screen has a dark background with a light gray card. The left side, titled 'ChatSphere', describes the app's features. The right side, 'Create Account', includes a sub-header, a sign-up instruction, three input fields for full name, email, and password, a blue 'Sign Up' button, and a link for existing users.

ChatSphere

Create your account and start real-time chat with a clean Discord and WhatsApp inspired layout.

- Live online users
- Instant message updates
- Image messages, edit and delete support

Create Account

Sign up to continue.

Full Name

Email

Password

[Sign Up](#)

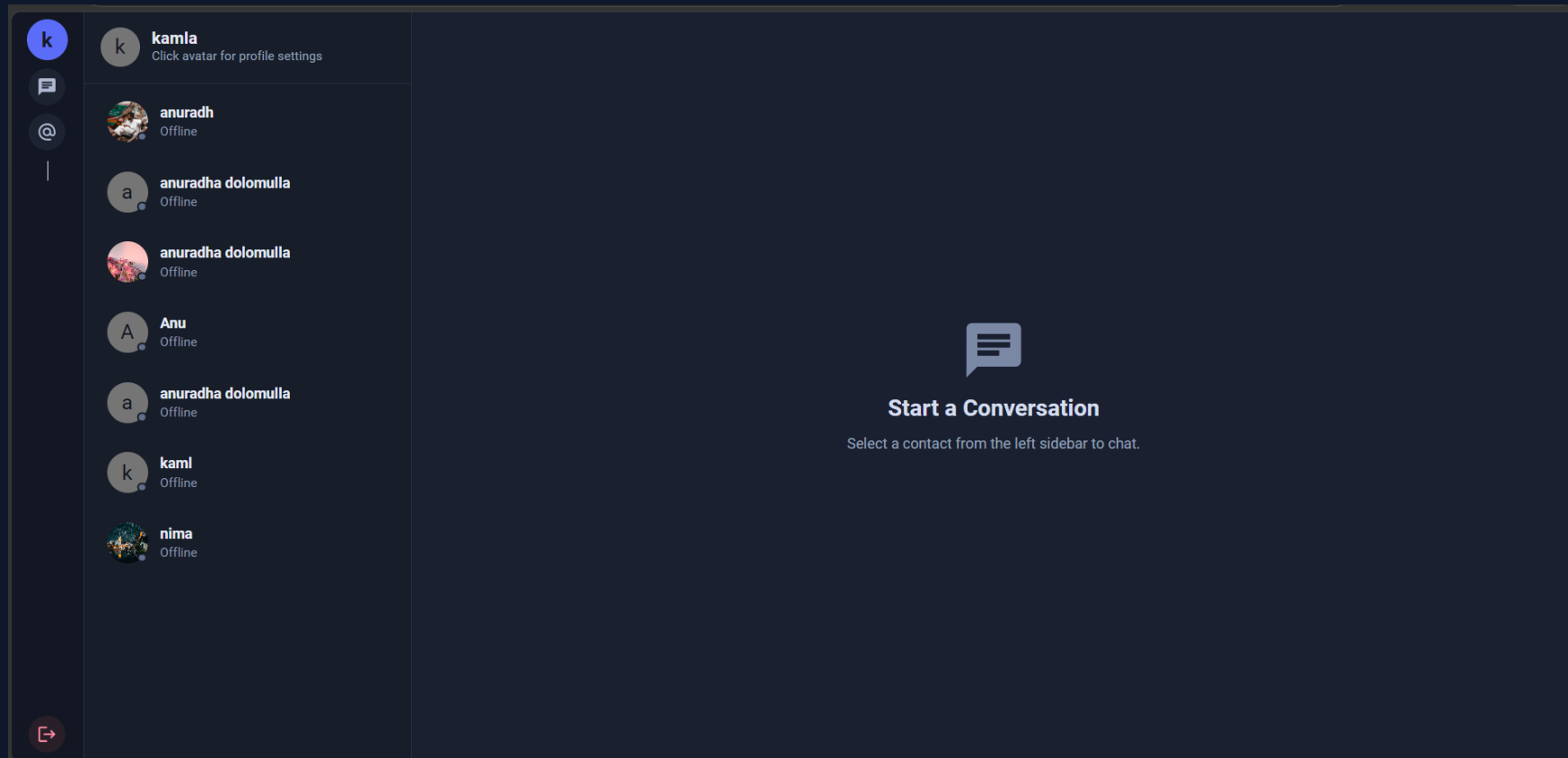
Already have an account? [Log in](#)

Login Screen

Users enter email/username and password to securely sign in. JWT authentication is used after successful login.

Sign Up Screen

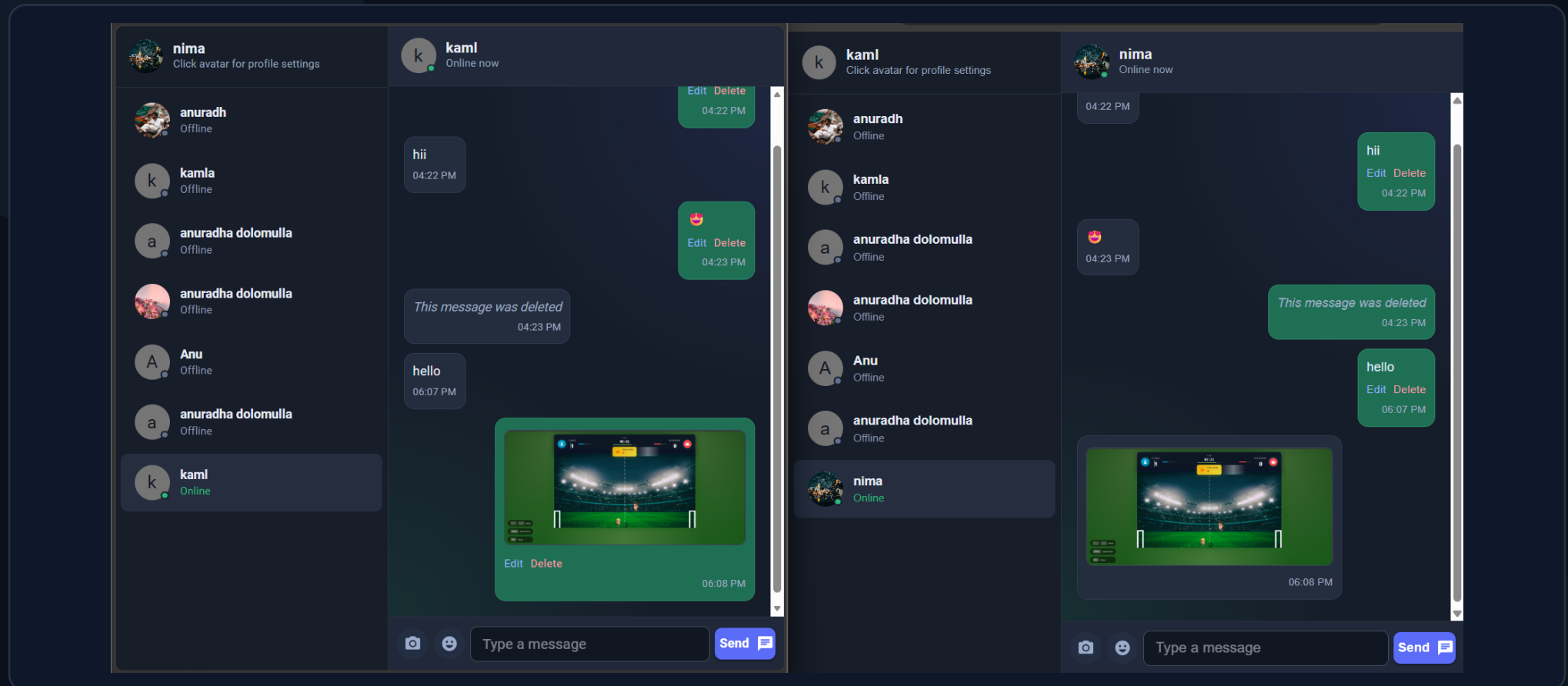
New users enter display name, username, email and password. After submission, email OTP verification completes the registration flow.



Chat Dashboard - Start Conversation

The main workspace shows users in a sidebar with online/offline status. The user selects a contact from the sidebar to start chatting.

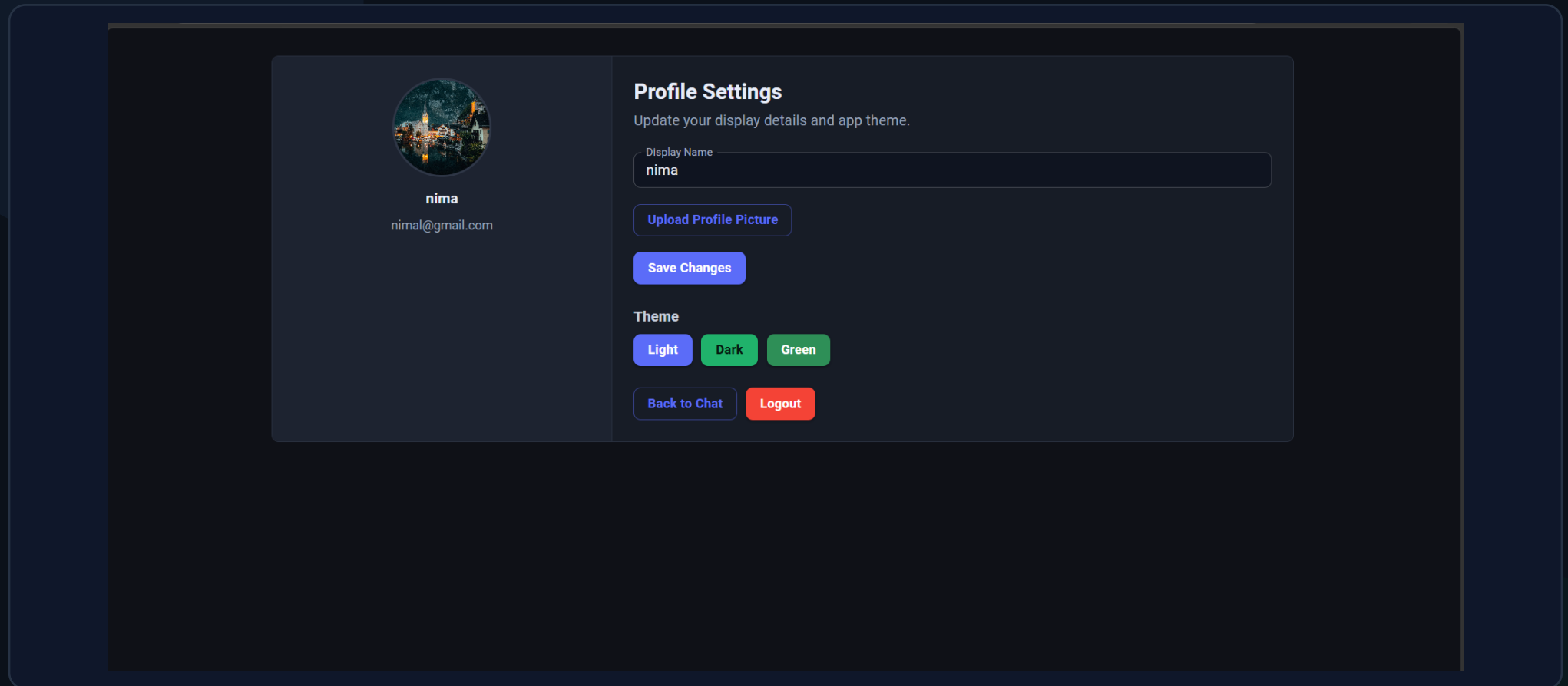
Key items: Contact list, User presence, Start conversation page



Real-Time Messaging Screen

This screen demonstrates live messaging between users. Messages appear instantly, media messages can be shared, and users can edit/delete messages when needed.

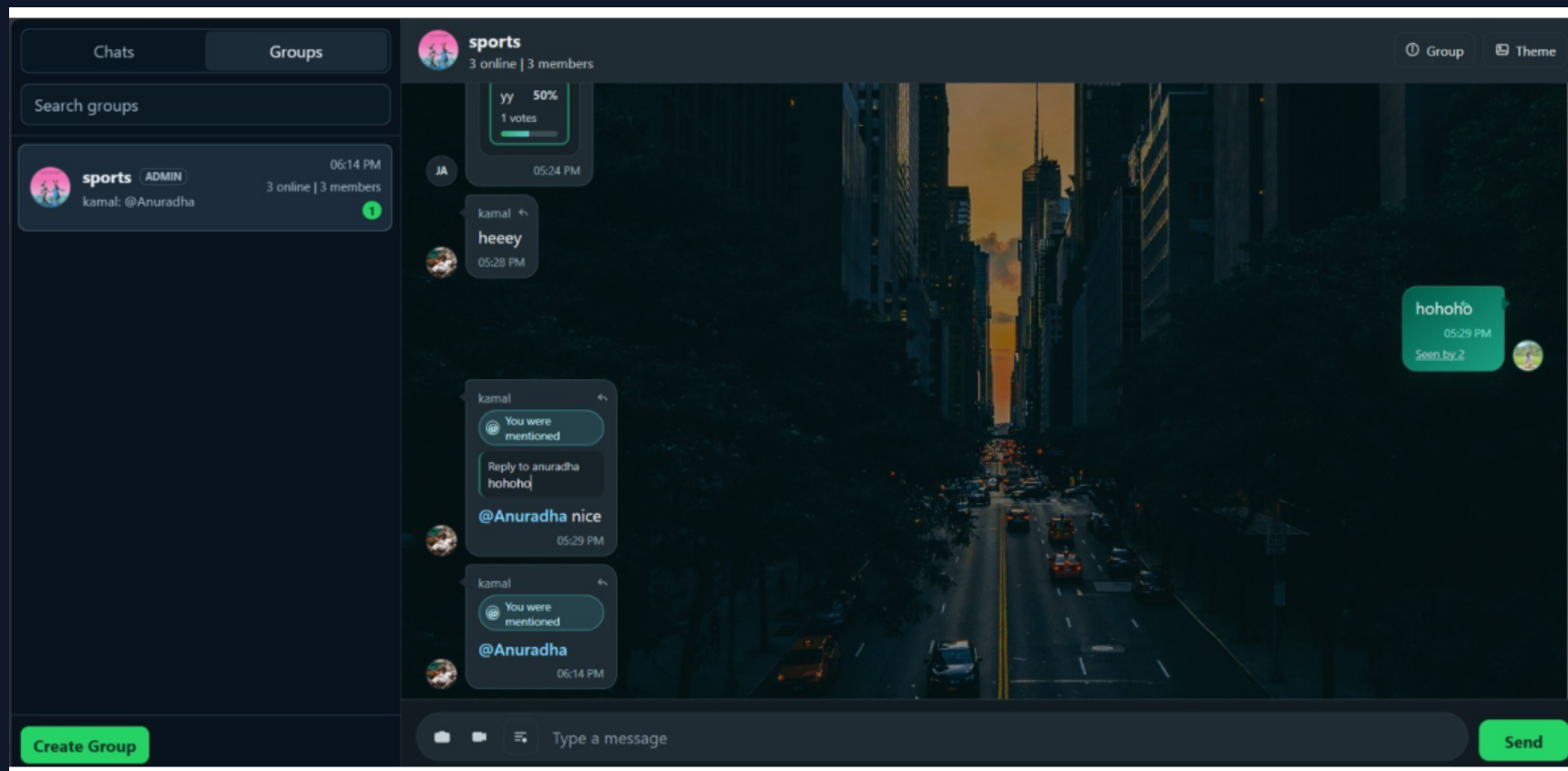
Key items: Real-time text messages, Image/media message, Edit/delete options, Online presence



Profile Settings and Theme Selection

Users can update profile details, upload a profile picture, select app themes and log out securely.

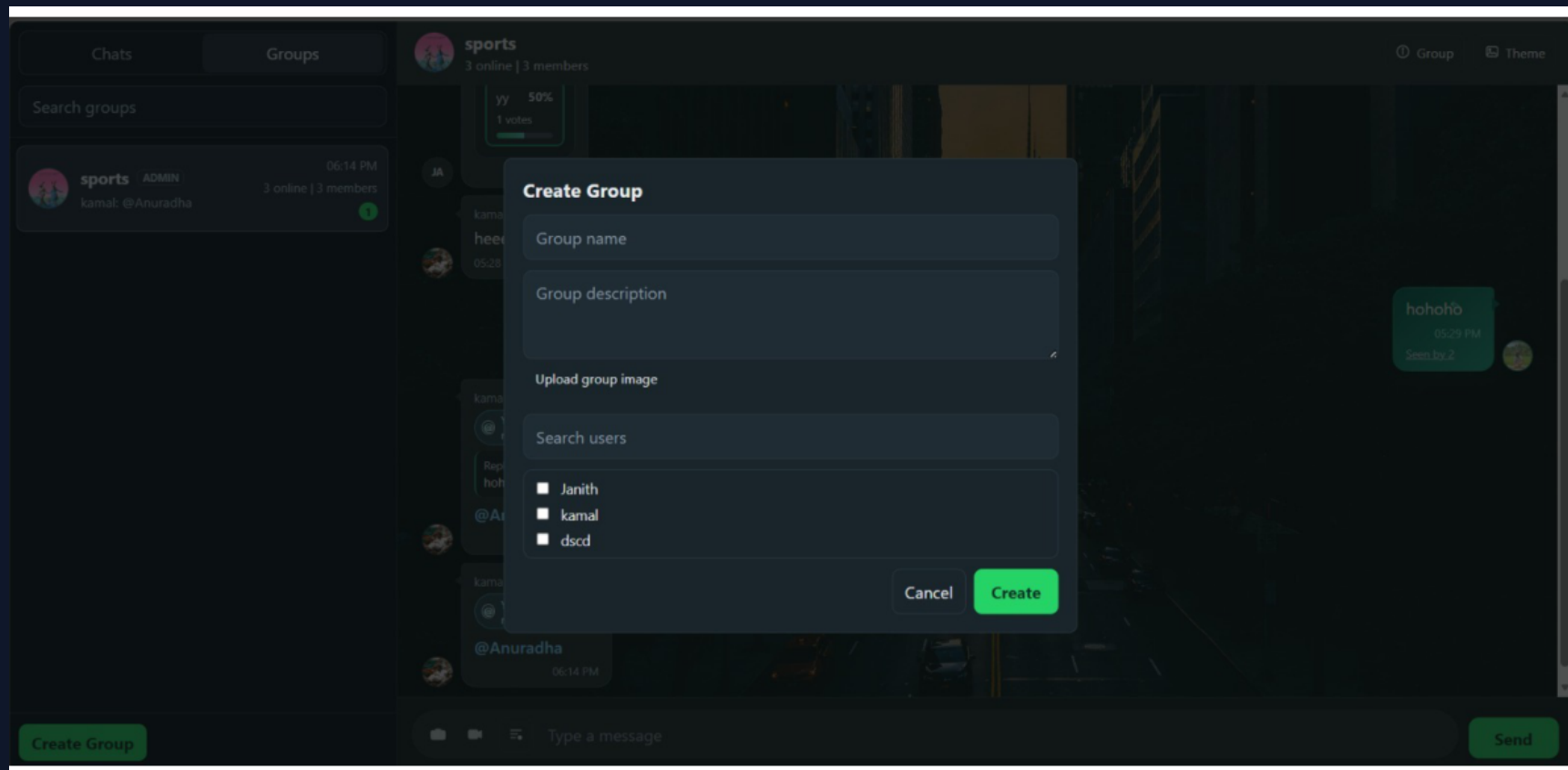
Key items: Profile image, Display name update, Theme selection, Logout



Group Chat Workspace

This screen shows the group chat workspace where users can communicate inside groups while still using the chat sidebar and real-time message area.

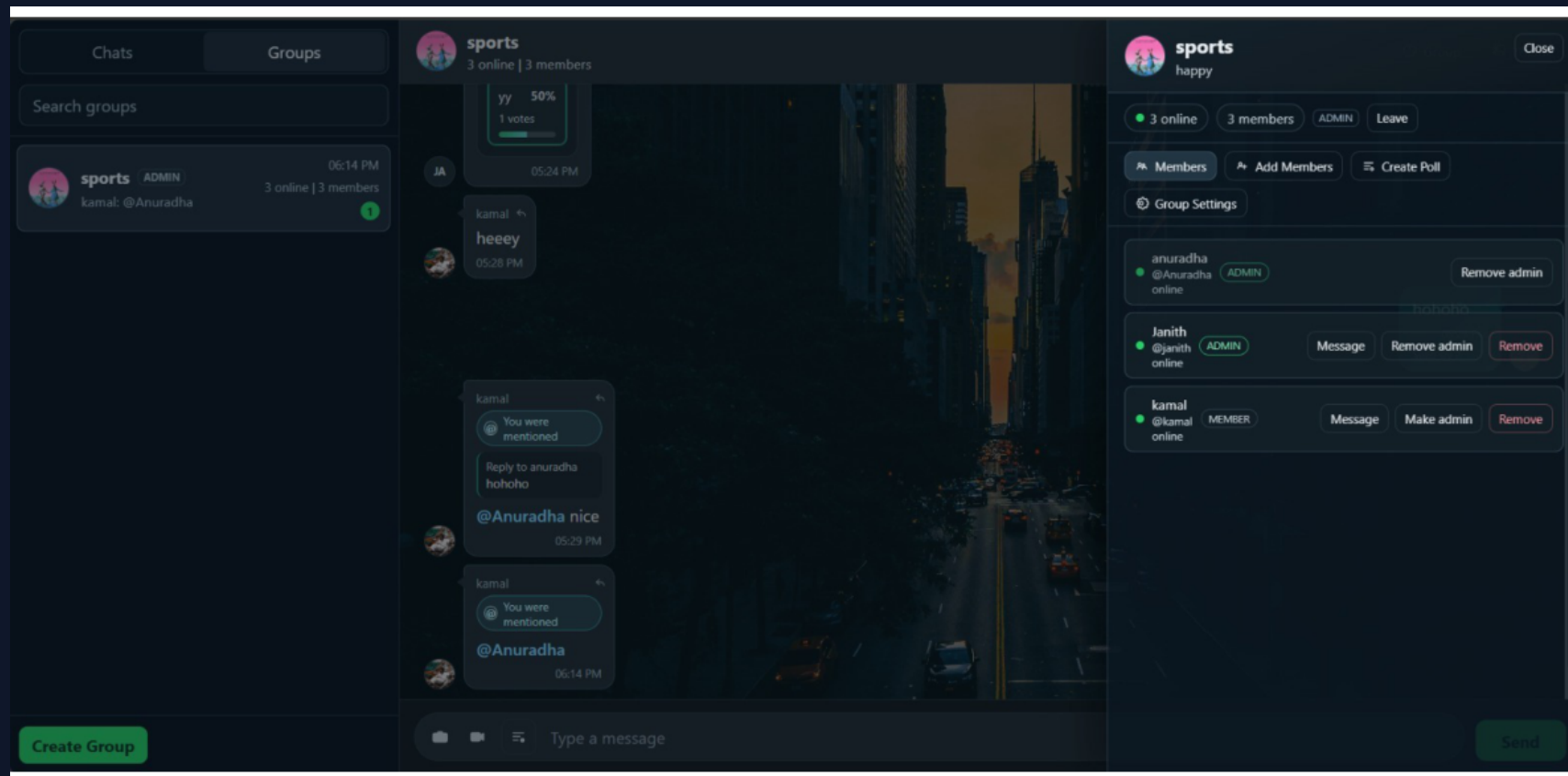
Key items: Group chat, Real-time updates, Sidebar navigation



Create Group Dialog

Users can create a new group, add a group name, choose members, and start a group conversation.

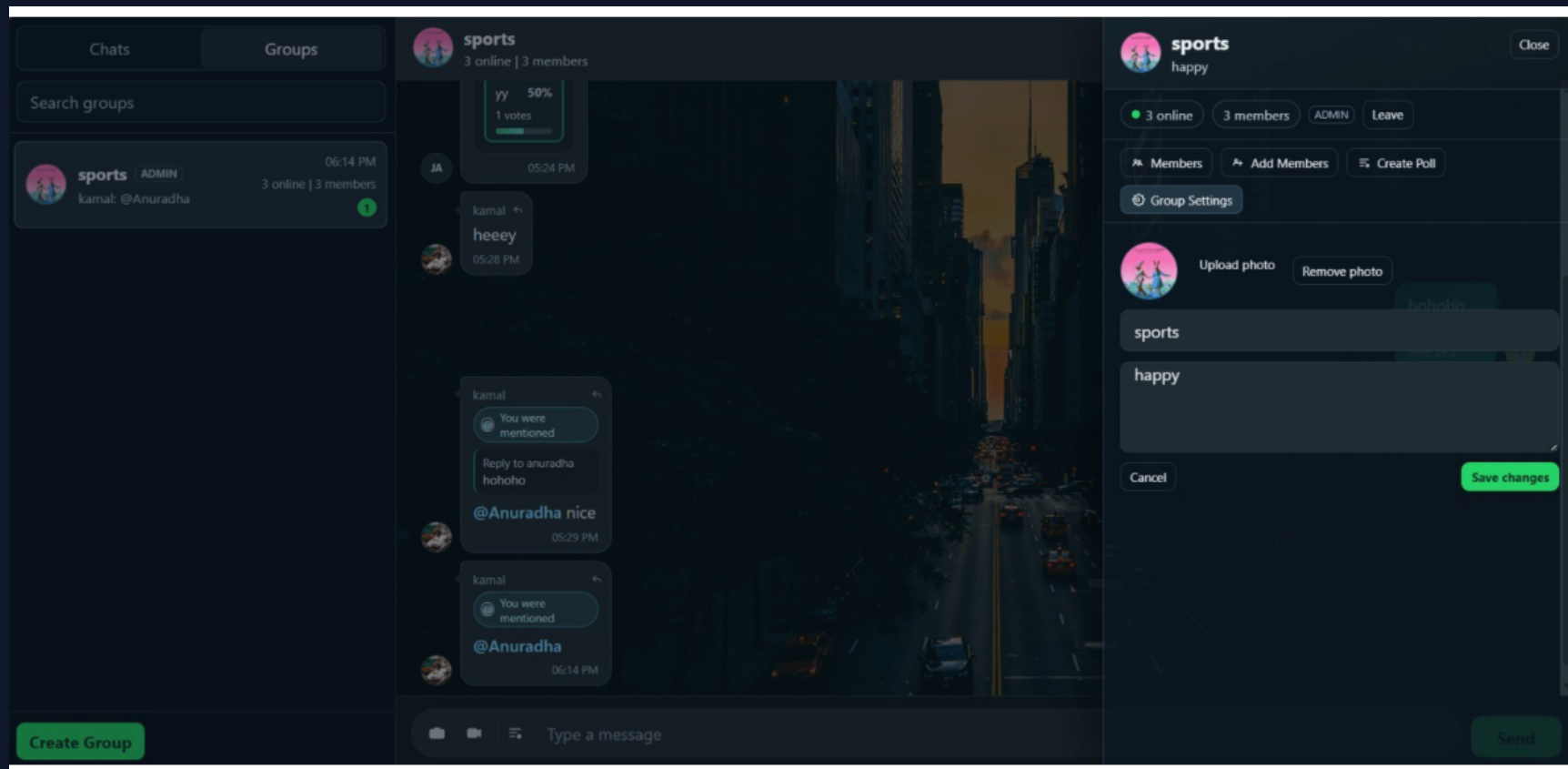
Key items: Create group, Member selection, Group setup



Group Members and Admin View

This view shows group members and admin controls. The creator or admin can manage group membership and roles.

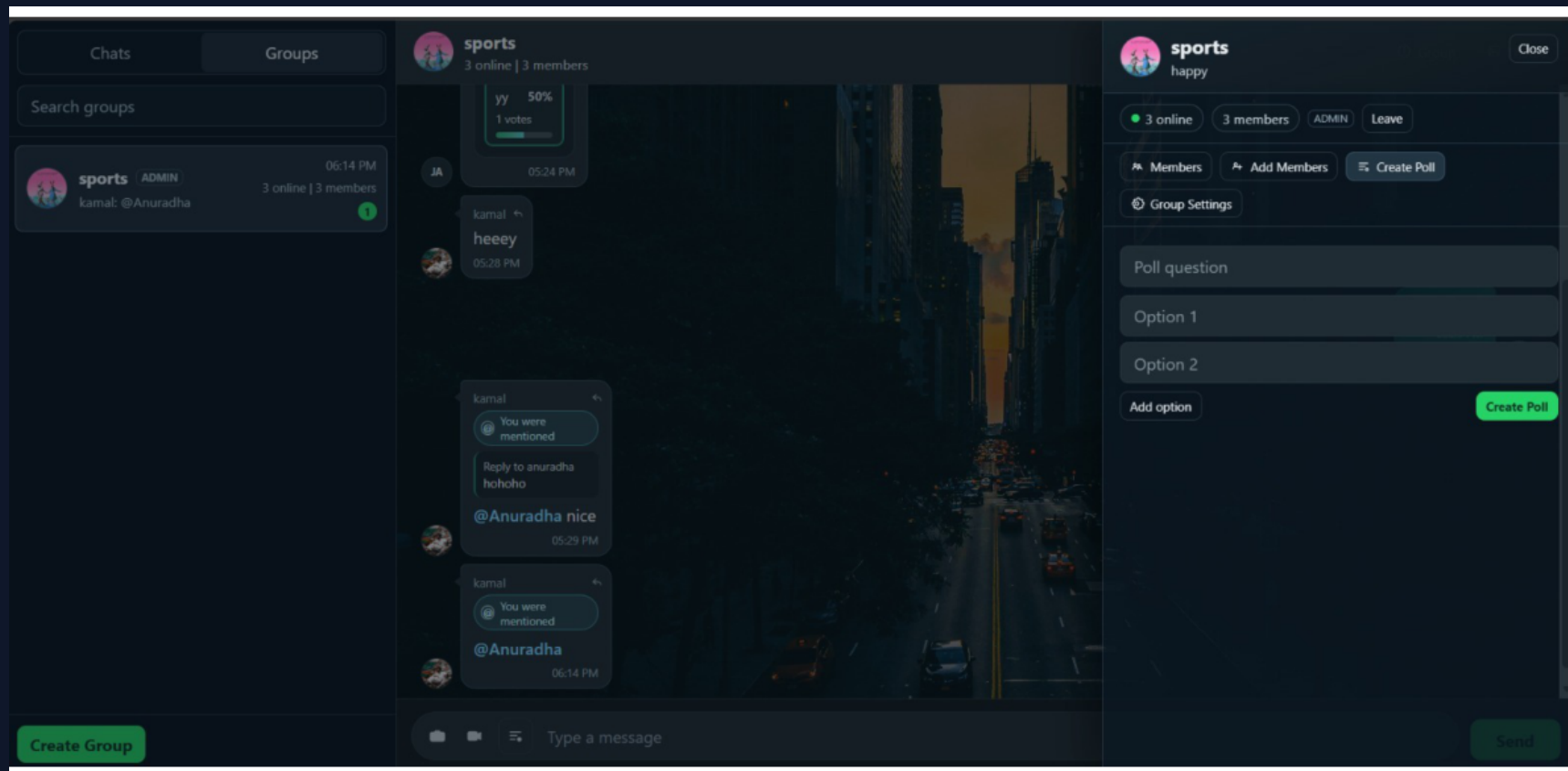
Key items: View members, Admin role, Member management



Group Settings and Management

Group settings allow admins to update group details, manage members, and control group actions.

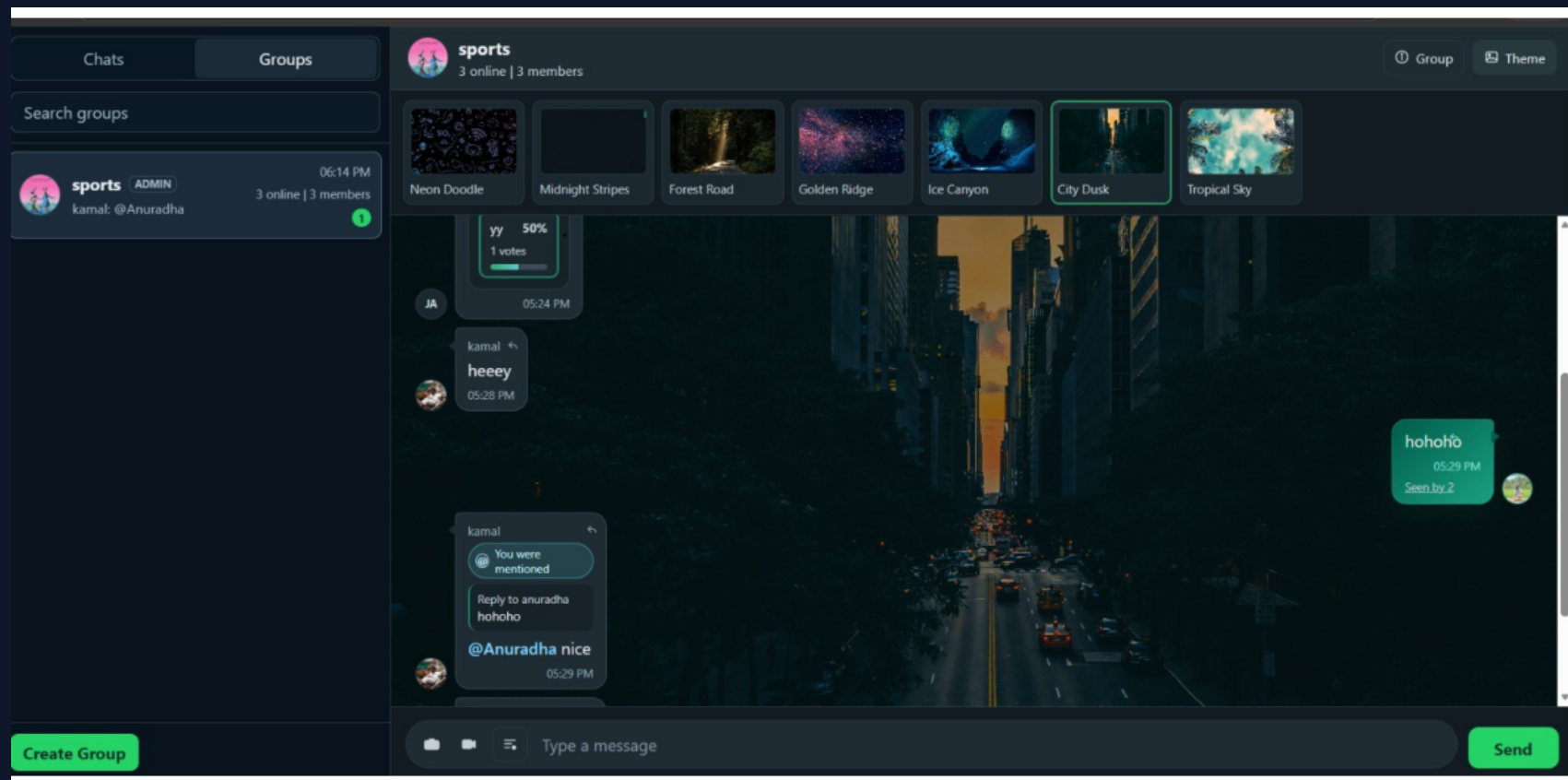
Key items: Group settings, Remove members, Admin controls



Poll Creation Feature

Users can create polls inside group chats to collect opinions from group members.

Key items: Create poll, Group collaboration, Poll options



Media Sharing in Chat

The chat interface supports rich message sharing including images, photos or other media inside conversations.

Key items: Media message, Image sharing, Chat history